

Hypothesis: A perfect day conveys internal time



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ABSTRACT

In 2007 the International Agency for Research on Cancer [IARC] concluded “shift work that involves circadian disruption is probably carcinogenic to humans” (Group 2A). To investigate the “probable” causal link, information on individual chronobiology is needed to specify exposures to circadian disruption associated with shift work. In epidemiological studies this information is usually assessed by questionnaire. The most widely used Morningness-Eveningness-Questionnaire (MEQ) and MunichChronoTypeQuestionnaire (MCTQ) reveal information on circadian type (MEQ) and actual sleep behaviour (MCTQ). As a further option we suggest to obtain preferred sleep times by using what we call the perfect day (PD) approach. We hypothesize that a PD – as a day of completely preferred sleep behaviour – captures pristine internal time. We argue that the PD approach may measure internal time more accurately than the MEQ and MCTQ which convey influences by work and social time pressures. The PD approach may therefore reduce misclassifications of internal time and reveal circadian disruption caused by different shift systems.

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Introduction

In 2007 the International Agency for Research on Cancer (IARC) concluded that shift work involving circadian disruption is probably carcinogenic to humans (Group 2A) [1]. Since then, several studies have examined associations between shift work and cancer. Surprisingly, most investigations have been based on the assumption that working at the civil night has the same biological effect on all humans. However, this assumes that internal time [2] and external time [3] (see Table 1 with glossary) are equivalent. Importantly, humans' internal times vary inter-individually (e.g. morning persons have an earlier cycle, evening persons have a later cycle) and intra-individually (e.g. changes by season or age). This means that ignoring chronobiological diversity can bias effect estimates of the same work shifts on chronobiologically different individuals, leading to under- or overestimations of possibly associated risks. Hence, an accurate assessment of exposures to – or doses of – chronodisruption (operationalized as the split physiological nexus of internal and external times [4]), is needed to explore the nature of associations between shift work and cancer. To this end, accurate information on the individual internal time is a *conditio sine qua non*.

The key question is: How can we assess individuals' internal time in epidemiological studies? To this end, we offer a novel approach via the following steps:

1. We explain why measuring internal time is a principal challenge.
2. We summarize how the widely used Morningness-Eveningness-Questionnaire [MEQ] and MunichChronoTypeQuestionnaire [MCTQ] have been employed to assess internal time in epidemiological studies.
3. We identify limitations of using either the MEQ or the MCTQ.
4. To assess pristine internal time we hypothesize that “A perfect day conveys internal time”. We describe how the perfect day [PD] approach uses preferred sleep behaviour to approximate internal time and explain advantages and limitations.
5. Conclusions with perspective considerations close the paper.

Measuring internal time is a principal challenge

To estimate internal time (IT) is demanding as it depends on the individual chronotype which (co-)determines an individual biological day – with the propensity of activity – and an individual *biological night* – with the propensity to be asleep [2]. However, actual wake/sleep behaviour is influenced by various external factors which act as *zeitgebers*. These external influences complicate deriving an individual's pristine internal time, i.e., the internal time

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Table 1
Glossary.

Variables	Definitions
Biological Day	Individuals' propensity to be awake; see chronotype and internal time
Biological night	Individuals' propensity to be asleep; see chronotype and internal time
Chronotype	Genetically (co-)determined [13,14] "temporal phenotype of an organism" [15]; see internal time [6]
External Time	Environmental time determined by sun, occupational and social times [3]
Free Day Shift System	Predominantly determined by individuals (such as family and friends), social events (such as cultural events) or private duties (such as house work) in addition to natural and occupational zeitgebers
Internal Time	Biological time as individuals' genetically (co-) determined [13,14] propensity to be asleep (biological night) and awake (biological day); see chronotype
Natural Shift System	Determined by natural, environmental zeitgebers only (such as sun light, temperature or natural noises)
Perfect day	A day with preferred sleep times within two weeks entirely for oneself
Pristine Internal Time	Internal time without external influences such as work during an individual's biological night
Work Day Shift System	Predominantly determined by the work schedule, and furthermore influenced by natural and social zeitgebers
Zeitgeber	External factors that synchronize the individual's biological rhythm with the environment; first defined by Aschoff [5] there are different zeitgebers that can be grouped by meteorological (such as sunlight and darkness, temperature or humidity) or ecological/social (such as other species' individual rhythms, noise or food availability)

without external influences such as work during an individual's biological night, from the actual sleeping behaviour: On work days, humans' sleep behaviour is often obviously influenced by the work schedule. But furthermore, various social *zeitgeber* pressures affect the sleep timing on work days and work-free days (hereinafter referred to as "free days"), e.g. the partner's chronotype, TV-events at particular points of time or social duties. The relevance of social *zeitgebers* was initially recognized by Aschoff in 1954, when he described the evening activity of salmon that was associated with the circadian preference of insects as their favorite food. Moreover, Aschoff reported that male rats adapt to female rats' activity rhythms when raised next to each other [5]. Arlie Hochschild described a Second Shift covering work outside the official working hours including housework and childcare. For women in the 1960s and 1970s she described an extra month of twenty-four-hour days (per year) working in this Second Shift in addition to the regular First Shift at the contracted workplace [6]. Taken together, social influences are highly relevant for the individual sleep behaviour.

Various, often co-existing *zeitgebers*, bond to different shift systems: On work days occupational *zeitgebers*, like the work schedules, have a main influence on activity and sleep behaviour (*Work Day Shift System*) whereas on free days (*Free Day Shift System*) social life requirements will have major impacts as *zeitgebers*. In principle, separating free days and work days could be a feasible approach to assess work's impact on activity and sleep behaviour. But it may be misleading to derive the internal time from the Free Day Shift System. In addition to the social life requirements a worker's sleep behaviour on a free day likely remains affected e.g. by the work schedule of previous days. How much work schedules influence a worker's sleep behaviour will depend on shift details or individual preferences. Empirically, it takes our bodies time to overcome time-zone travel-associated jetlag. Similarly, workers will need time to adjust temporal wake-sleep habits to free day obligations or liberties. Hence, depending on shift and chronobiological details, the time span of a week-end away from

work may be too short to wash out effects on their chronobiology accumulated over preceding work days. In summary, a variety of external influences may make it difficult to assess an individual's pristine internal time simply from his behaviour on, for instance, two free days. By way of contrast, isolating an individual from the influences of both work and social life would leave only exposures to natural *zeitgebers* – such as sunlight or natural temperature. This situation could be conceptualized as the *Natural Shift System*. This Natural Shift System may mirror pristine internal time as it includes natural *zeitgeber* information only (Fig. 1).

The MEQ and MCTQ in epidemiological studies

Epidemiologists regularly use various questionnaires to reveal chronobiological information. Amongst these, the Morningness Eveningness Questionnaire and the Munich Chronotype Questionnaire are the two "most widely used tools in epidemiological research to assess circadian preferences" [7].

The Morningness Eveningness Questionnaire (MEQ)

In 1976, Horne and Östberg proposed the Morningness Eveningness Questionnaire (MEQ) [8]. It is described as a validated tool and considered as the gold standard measure of morningness [9]:

The MEQ contains 19 questions on individual preferences for physical or mental activity, sleep timing, morning alertness, appetite or tiredness at specific times of the day and self-assessing circadian preference. Horne and Östberg chose different scenarios for their questions: 9 questions investigate hypothetical situations (e.g. days to be planned entirely free), 9 further questions focus on real experiences (e.g. feeling best at time of day) and in one question a comparison with others (self-assessing the diurnal preference) is used. The questionnaire consists of multiple choice questions and ordinal scales with responses being matched to different scores. The cumulative result of these scores is classified into one of five categories from definitively evening type to definitively morning type. Different definite cutoff points for the circadian preferences' categories are discussed in the literature [9]. Overall, the MEQ provides a cumulative result that enables to assess circadian preference as categorical variable.

The Munich Chronotype Questionnaire (MCTQ)

The MCTQ, first described by Roenneberg et al. in 2003, is a questionnaire to assess information on individual circadian timing [3]. In its current form, the questionnaire includes 14 questions [10] that focus on individual sleep behaviour specified for work and free days – with free days being defined as normal circumstances on free days without partying. Hence, the MCTQ collects information on current real life situations. It can be used to show quantitative differences in sleep behaviour between work and free days. One outcome, the Social Jet-Lag as the absolute difference between mid-sleep time on work and free days [11], can be calculated in fractions of hours. The mid sleep time on free days is furthermore used to assess individual chronotypes. Assuming a normal distribution of chronotypes, different categories of chronotypes are assigned by splitting the continuous results obtained from the investigated population according to quartiles or deciles or other cutpoints [11]. In addition to information on sleep timing, the MCTQ can also be used to derive information on sleep duration. In 2013, the MCTQ_{shift} extended the MCTQ to shift workers [12]. It uses the same questions as the MCTQ but focuses on different days within the duty roster. It derives the chronotype by the midsleep time of free days after evening shifts. Hence, the methodology of the MCTQ and the MCTQ_{shift} to assess internal time is comparable.

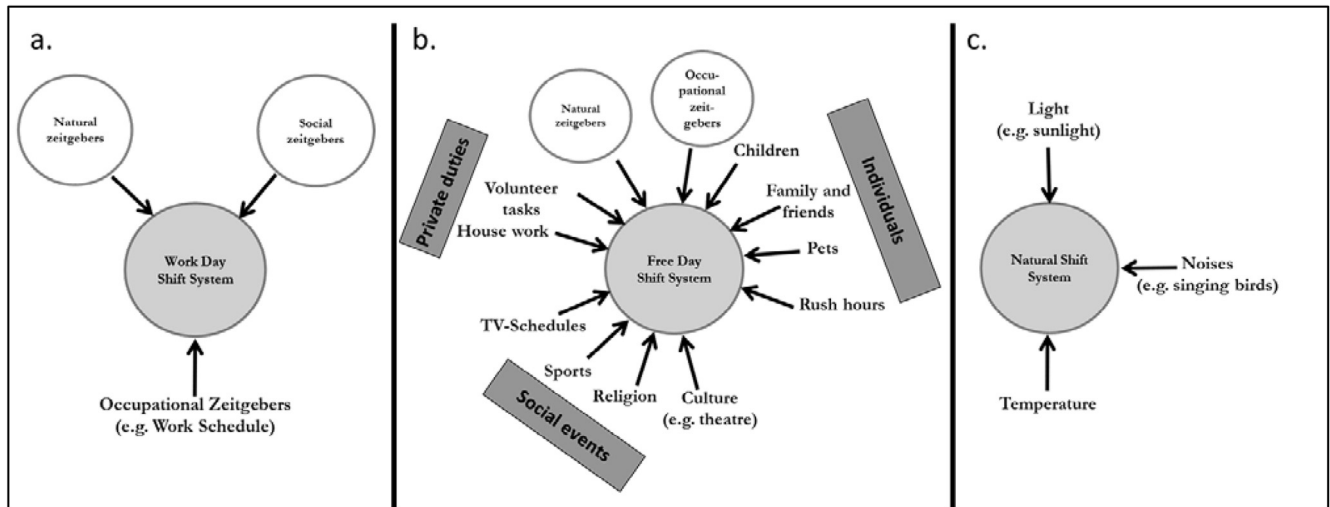


Fig. 1. External influences on different shift systems. a) The Work Day Shift System is predominantly determined by the work schedule, and is furthermore influenced by natural and social zeitgebers. b) The Free Day Shift System is determined by individuals (such as family and friends), social events (such as cultural events) or private duties (such as house work) in addition to natural and occupational zeitgebers. c) The Natural Shift System is determined by natural, environmental zeitgebers only (such as sun light, temperature or natural noises).

Overall, both questionnaires provide continuous variables to quantitatively describe sleep behaviour on free and work days.

Limitations of the MEQ and MCTQ

The MEQ and the MCTQ both provide important information on individual chronobiology. However, there are limits when assessing biological day and night via either approach.

The MEQ combines preferred times with real-time scenarios and may therefore be close to information on the pristine internal time but the cumulative result of its 19 questions does not give specific information on the individual biological day and night. However, to assess individual internal time, the MEQ's first two questions could be used: "Considering only your own "feeling best" rhythm, at what time would you get up if you were entirely free to plan your day?" and "Considering only your own "feeling best" rhythm, at what time would you go to bed if you were entirely free to plan your day?". However, these questions reveal information on a preferred time "to go to bed" and "to get up". Unfortunately, this information does not necessarily provide us with the actual sleeping times which would be of interest to assess internal time. Let us give an example:

Imagine a person who loves to read books while staying in bed. Hence, this person may answer that on a day of his preference he would go to bed at 21:00h, but in fact he would stay awake until midnight while reading a fascinating crime story. Another person may love to have breakfast in bed, so he may answer to leave the bed at 10:00h, even though he did sleep until 8:30 h only.

Therefore, asking for preferred times to go to bed and to get up when individuals were entirely free to plan their day, does not necessarily reveal information on their actual sleep times. Hence, these questions do not provide precise information on the pristine internal time.

In contrast, the MCTQ asks about the actual sleep times, and information regarding free days is used as reference for the internal time. Hence, information is captured for both the Work Day Shift System and the Free Day Shift System. Regrettably, there is no way to disentangle the Natural Shift System or pristine internal time. To use the sleep behaviour on free days as surrogate for the internal time could be misleading, as it may underestimate the

influence of the social requirements on sleep behaviour and it ignores the influence of previous work days. To give two examples:

First, imagine a cook with a late chronotype who regularly works in the evenings. On work days he returns home after midnight. In order to enable him to get enough sleep, his wife cares for the children to send them to school early in the morning. On his free days – which are not during the week-end – he is responsible for sending the children to school, so he gets up early. For this cook, the sleep behaviour on free days is actually more out of phase with his pristine internal time than on work days.

Second, imagine a baker who is an early chronotype. He has to wake up early in the morning to start working. He may be closer to his preferred sleep times during his work days than on free days when his family and friends expect him to spend time with them during the evenings and don't accept his work-associated early bedtimes.

In both examples, using the sleep behaviour on free days to assess internal time would be misleading.

Hypothesis: a perfect day conveys internal time

Definition

In order to identify individuals' pristine internal time (or the Natural Shift System) we suggest combining information on preference (as used in the MEQ) and on specific sleep times (as assessed in the MCTQ) by asking individuals about their 'perfect day' (PD). We define the PD as a day with preferred sleep times without external social and work influences on sleep behaviour:

Imagine you have two weeks just for yourself, there is no other person influencing your day, no animal that has to be cared for at specific times, no work, no appointments, you can schedule all events and TV programs just the way you want to. At the end of these two weeks, what do you think your perfect day would be like? What time would be perfect to fall asleep and at what time would you feel best when waking up without using an alarm clock?

To assess information on sleep behaviour during perfect days, we would ask the respondent to complete the following sentences:

On a perfect day:

- I fall asleep ato'clock.
- I wake up ato'clock.

For the PD approach the information on the times of falling asleep and waking up would be relevant. However, it may be useful to ask for the time to go to bed and getting up, too, as these questions outline the situation and may facilitate imagining of this scenario. Hence, studies to validate the PD approach should investigate whether including these additional questions would be useful.

Discussion and evaluation

Advantages of the PD approach

The most important advantage of the PD approach may be that, depending on the individual ability of imagination and self-assessment, the PD reveals circadian information without external, non-natural influences. Therefore, the PD reveals information to zero in on the pristine internal time and the Natural Shift System. Compared to the first two questions of the MEQ which focus on the preferred times to “go to bed” and “to get up”, the PD’s precise wording avoids misclassification by focusing on the preferred time window of sleep. Furthermore, it diminishes the social impact that may be a part of a preferred day, by explicitly excluding influences through individuals and offering the ability to schedule all events at the pristine personal discretion. Compared to the MCTQ, the PD may capture internal time without the individual’s social life influence during free days.

How the three approaches (MEQ, MCTQ and PD) to assess internal time differ and how imprecise wording and ignoring external influences may yield significantly misclassified internal times is illustrated in the following example:

Imagine an opera singer, who usually performs during the evenings. Hence, on work days she goes to bed late at night, i.e., at

03:00, and directly falls asleep. She sleeps for 6 h only, as she has to attend rehearsals in the morning at 10:30. On free days she works for charity by preparing breakfast for schoolchildren and therefore she has to wake up at 06:00. As getting up early makes her tired in the early evening, she goes to bed at 22:00h and immediately falls asleep. On a preferred day or on a perfect day when she could choose her sleep times completely on her own preference, she would wake at 9:00, meditate for an hour, and then get up at 10:00h. As she loves listening to classical music while lying in bed, during her preferred days or perfect days she would go to bed at 23:00h and stay awake to listen to music for another two hours before sleeping.

How this example’s details translate differentially into internal time when using the different approaches is presented in Fig. 2.

Another advantage of the PD approach is that it enables us to assess chronodisruption as a cumulative exposure variable over time – for instance the total chronodisruption experienced by working fixed hours which do not overlap with internal time over years or decades. A continuous measure of chronodisruption may be used to assess limiting values or dosage-dependent risks. To this end, chronodisruption may be computed [2] by comparing shift times – as external time – with temporal information about the biological night. In practice, activity-rest patterns of internal and external times are compared (CD-hours; see Fig. 3). Assuming that the PD reveals information to approximate the pristine internal time, its addition to information about shift schedules or sleep behaviour on work days provides the necessary parameter to calculate individual chronodisruption hours during work and free days. Over years and decades, this approach allows computing a cumulative exposure of chronodisruption hours. As CD hours would result from both activity during the biological night as well as sleep during the biological day, it may be useful to distinguish between CD during the biological night and day.

Limitations of the PD approach

The PD approach, too, suffers from limitations: We assume that the PD approach provides information about pristine internal time and the natural shift system and that the latter is the most physiological shift system. Whether this assumption is valid remains open at this stage.

Moreover, the ability to imagine a PD may vary between individuals and may be influenced by e.g. the present alertness, time

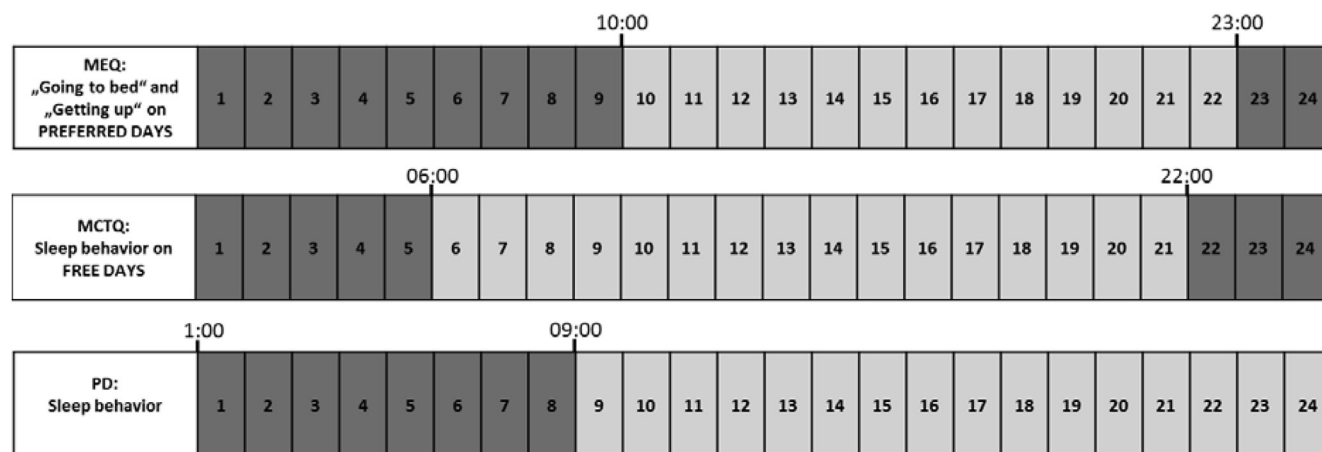


Fig. 2. Assessment of internal time based on the a) MEQ (the first two questions asking for times to “go to bed” and “to get up” when individuals “were entirely free to plan” their day), b) MCTQ (referring to the sleep behaviour on free days) and c) the sleep behaviour imagined or self-assessed for PDs. The hours of sleep are presented in dark grey, hours of activity in light grey.

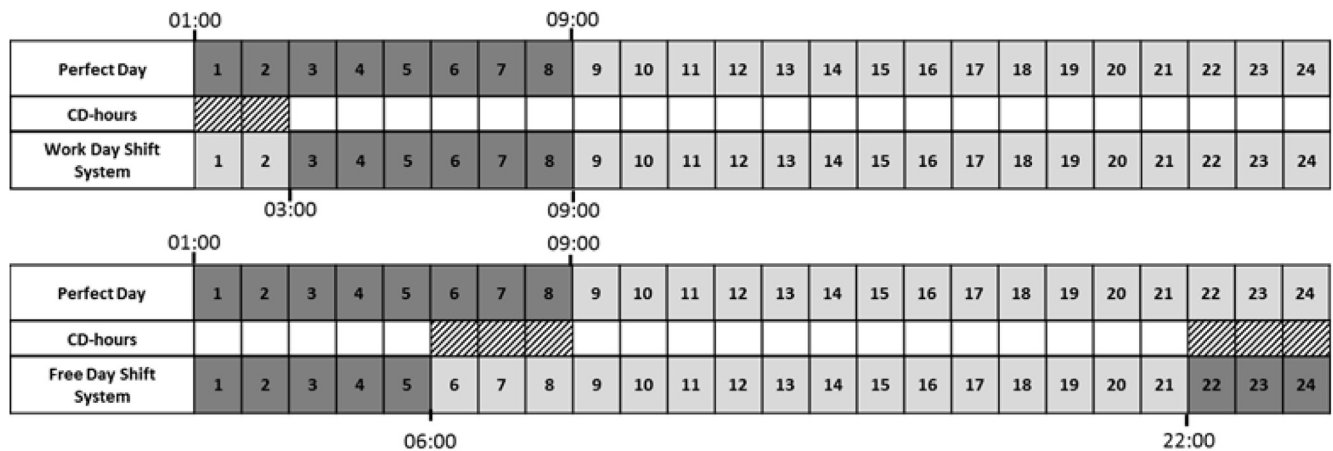


Fig. 3. Assessing CD hours by comparing the opera singer's [see text for details] sleep behaviour on PDs with Work Day Shift Systems or Free Day Shift Systems. Hours of sleeping are marked in dark grey, time periods of activity are marked in light grey, and CD-hours (as the overlap of opposite conditions) as dashed.

of day or season. In addition, how we propose to assess a PD has not yet been empirically tested. Another issue is that if persons had not had a PD for years or decades, they may remember a comparable situation that happened a long time ago. As the internal time seems to be influenced by age [11] they may assume the sleep behaviour of their younger age to be still perfect even though they are older. This may lead to misclassification bias.

The shift system on work days is always a mixture of social, occupational and environmental zeitgebers. Therefore, contributions of the Work Day Shift System to circadian disruption cannot be attributed to facets of work alone.

Finally, our conceptualization of three shift systems (Work Day Shift System, Free Day Shift System and Natural Shift System) may be a simplification. Ultimately, therefore, it may be valuable to explore how specific external factors impact on individual sleep behaviour and circadian disruption.

Conclusions

Assessing internal time to investigate occupational and social impacts on circadian disruption is a challenge. The MEQ and MCTQ are both important questionnaires to assess information on chronobiology, but disallow the assessment of pristine internal time as described in this paper. The proposed PD scenario could provide a remedy. The PD may enable us to estimate the pristine internal time, as it focuses on specific sleep behaviour, leaving out as much as imaginable, occupational and social influences. If proven valid, the PD could reduce misclassification of individuals' internal time. However, high quality epidemiological studies are needed to investigate whether and how the PD-approach – which we described in theory – will be useful in practice.

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